

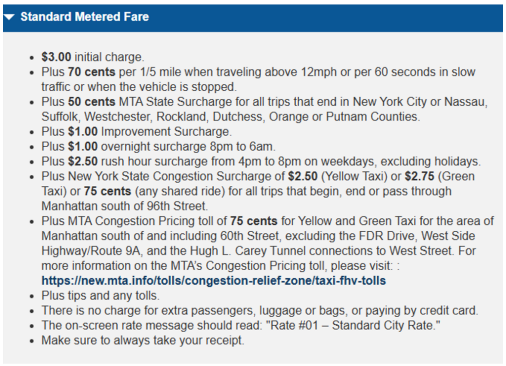
Newyork Taxi data Analyses: what is the price ?

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Introduction

Welcome to New york city, a city of dreams, life stories, and taxi's!
If one decides to try out taking a taxi in New york, they might lookup the pricing of taxi journeys online. On the official website one can find this text:



Standard Metered Fare

- **$3.00** initial charge.
- Plus **70 cents** per 1/5 mile when traveling above 12mph or per 60 seconds in slow traffic or when the vehicle is stopped.
- Plus **50 cents** MTA State Surcharge for all trips that end in New York City or Nassau, Suffolk, Westchester, Rockland, Dutchess, Orange or Putnam Counties.
- Plus **$1.00** Improvement Surcharge.
- Plus **$1.00** overnight surcharge 8pm to 6am.
- Plus **$2.50** rush hour surcharge from 4pm to 8pm on weekdays, excluding holidays.
- Plus New York State Congestion Surcharge of **$2.50** (Yellow Taxi) or **$2.75** (Green Taxi) or **75 cents** (any shared ride) for all trips that begin, end or pass through Manhattan south of 96th Street.
- Plus MTA Congestion Pricing toll of **75 cents** for Yellow and Green Taxi for the area of Manhattan south of and including 60th Street, excluding the FDR Drive, West Side Highway/Route 9A, and the Hugh L. Carey Tunnel connections to West Street. For more information on the MTA's Congestion Pricing toll, please visit: : **https://new.mta.info/tolls/congestion-relief-zone/taxi-fhv-tolls**
- Plus tips and any tolls.
- There is no charge for extra passengers, luggage or bags, or paying by credit card.
- The on-screen rate message should read: "Rate #01 – Standard City Rate."
- Make sure to always take your receipt.

Figure 1: official pricing table used by New york taxi [1]

To sum up the text in Figure 1, a passenger pays $4.50 to get in the taxi, $2.75 for existing in a taxi in certain parts of Manhattan. In addition, a passenger has to pay $0.70 per 1/5 mile or per 60 seconds if traveling slower than 12 Mph. And Finally, during rush hour (4 pm-8 pm on weekdays) there is an extra charge of $2.50, and during the night (8 pm-6 am) there is an extra charge of $1.00

From the above, a question arises mostly focused on the rush hour commutes. Rush hour is $2.50 extra, but is also most likely to be more expensive per kilometer due to the $0.70 per minute if speed less than 12 Mph regulations. These are al assumptions but create the main research question:

When is it most expensive to take a taxi in New york, and by how much?

Another thought was that during winter, more people use transport rather than walking. This will lead to increased traffic making overall speeds slower, which results in a higher price for the same commute. This creates the secondary research question:

Does the price/kilometer depend on the month?

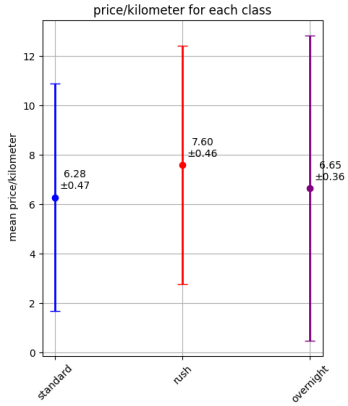
Data sample

For the analyses, data from new yorks green taxi's was used [2]. This data set consisted of more than 540000 commutes, each with date's, times, distance ,price and a whole range of other statistics. the relevant columns used are:

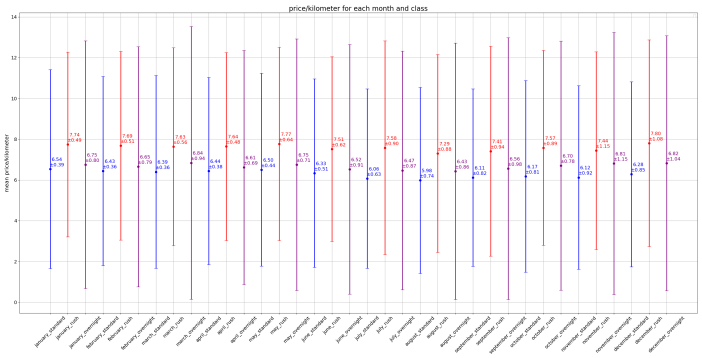
- date, date at which the commute started
- pickup time, time at which the commute started
- dropoff time, time at which the commute ended
- distance, The distance of the full commute
- kost, The amount to be paid by the passenger(s)

The data wass also filtered to remove instances of 0 kost, 0 distance, starting in 2024, overlapping classes (standard and rush hour for example) and price/kilometer greater than 100. After the filtering process, the data was separated into sub-samples of 100 commutes

Results and discussion



(a) price/kilometer for each class



(b) price/kilometer for each month and class

Both of the above plots show the calculated mean±uncertainty and error bars representing the standard-deviation

Figure a shows the price/kilometer for each class. From this figure, the main research question can be answered, namely that as expected rush hour commutes are the most expensive with an average additional charge of $ 1.30 per kilometer compared to standard commutes. Overnight commutes came out in the middle, charging an average additional charge of $ 0.37 per kilometer compared to standard commutes.

An interesting observation to make here is that the additional charges of $ 1.00 Overnight and $2.50 Rush hour are a different ration (1:2.5) than the average price differences (1:3.51). From this it can be concluded that other factors are contributing to the increased price of Rush hour. This is most likely to be (as said in the hypotheses) caused by slower traffic during Rush hour and the effect it has on the slower than 12Mph regulations.

Figure b shows the price/kilometer for each month and class. It is a little bit hard to tell from this one plot, which is why in the program, this plot is also split into 3 other plots showing al months, but only 1 class each. these plots are not included in this paper to save space.

Figure b answers the secondary research question. For al types, the average price / kilometer is lower around the summer period or otherwise said higher around the winter period. Rather this is actually caused by the increased traffic caused by the winter cold and poor weather could not be concluded from the analyses alone.

References

[1] Taxi fare TLC. *Taxi fare in New york city table* Available at: https://www.nyc.gov/site/tlc/passengers/taxi-fare.page.

[2] 2025 Green Taxi Trip Data. *NYC Green Taxi Trip Data 2025.* Available at: https://data.cityofnewyork.us/Transportation/2025-Green-Taxi-Trip-Data/nkud-f3se/about_data.